

# Cold-Chain Logistics, Last Mile Delivery\*

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## ABSTRACT

This report describes a project that focuses on addressing the challenges associated with the last mile delivery process, particularly the sustainability issues, complexity of the delivery process, and customer demands. The project aims to use the tools of the Fourth Industrial Revolution (4IR), such as the Internet-of-Things, software engineering, and machine learning, to develop and monitor the hardware of a cooler box designed to allow E-delivery vehicle manufacturers to enter the cold chain delivery business. The report includes a literature review on cold chain logistics and the use of Arduino sensors, as well as a detailed description of the hardware and tools used in the project, including a passive cooling container, temperature sensors, and a monitoring system built for smartphones. The report also discusses the procedure for product collection, which includes sensing the product temperature before inserting it inside the cooler box and canceling orders that do not meet the desired temperature range. Finally, the report describes the dynamic routing system that generates the shortest path to reach the destination.

## KEYWORDS

Logistics, Internet of Things (IOT), Environment Sensors, Passive Cooling, NFC, wireless communication, Cold-Chain, Fourth Industrial Revolution, Arduino

### 1. INTRODUCTION

Logistics is a subset of a supply chain management that coordinates the efficient flow and storage of goods and services between the source origin and consumers [1]. Throughout the past years the logistics industry has increased in its market value, with expected global hike of 13.71 trillion Euros by the year 2027, see Figure 1. The ongoing increase in the logistics industry is due to factors such as an increasing economic growth, the need for technology in developing countries, and the increasing population around the globe that results in an increased demand for food supply. This paper focuses on research in the discipline of logistics known as cold chain logistics which is the result of the maturation of the global logistics industry research.

Size of the global logistics industry from 2018 to 2021, with for (in trillion euros)

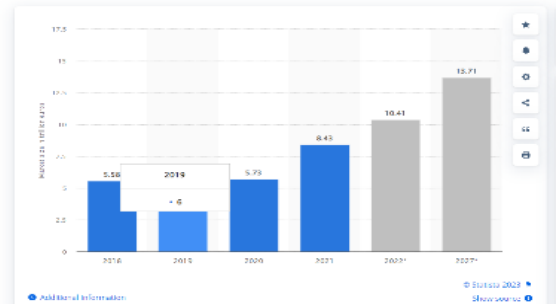


Figure 1 Logistics Industry market price

According to [2] cold chain logistics is described as a systematic project based on refrigeration technology. It is designed to be suitable for products that require to be kept in low temperatures throughout their lifecycle from their production, storage, transportation, and sales. The cold chain network ensures that the product's quality is maintained and minimizes product loss. These product demands raise high requirements for the cold chain network to function optimally. The hardware tools needed include refrigerators, passive cooling containers, temperature sensors, humidity sensors and light sensors. The software tools are crucial as they integrate all the hardware components together to develop the cold chain system. Although there are many software tools for cold chain development across different organizations, no complete and independent cold chain logistics system has been established so far [3].

For this report, we shall refer to the cold-chain system as the last-mile. The focus of the project (last-mile) lies on the development and monitoring of the hardware of a cooler box that is designed to allow E-delivery vehicle manufacturers to enter the cold chain delivery business. The Last-Mile aims to monitor automatic driver dispatching, package collection, dynamic routing, and the flexible delivery of products transported by battery powered cooler boxes or passive cooler boxes which rely on the ice stored within the cooler box.

### 2. LITERATURE REVIEW

In this section we analyze previous reports that focus on logistics, cold-chain logistics, traceability in cold-chain logistics and sensors. Yu Zhang in his research analyzed how the cold chain system works by focusing on [4]:

Refrigerated storage and processing tag. This is when the product is entered into the system. He mentions the use of barcodes, such

as the Radio Frequency Identification Device (RFID). The RFID is used on global scale as it is very efficient for large shipments tracking and the prevention of theft activities [4]. The other alternative is Near Field Communication (NFC), which is more flexible as many smartphone devices support it.

The sales and transportation stage, occurs once the products have been recorded. They are sent to the transport vehicle where they are re-scanned together with the vehicle registration. After the product is re-scanned, it is now stored in the system. At this stage the product will be monitored by all authorized parties at any time of delivery [4].

Lastly the marketing strategy; Yu Zhang mentions that the cold chain has the advantage of differentiated market because it guarantees low temperatures which will ensure product quality for the consumers.

Many developed countries such as China and USA have developed cold chain systems [4]. We analyze the results of the cold chain system of Liaoning Province in China made by Xiaopeng Zhao [5] in his research, which focused on the demand of fresh agricultural products and its impact on cold-chain logistics.

Table 2. Cold chain logistics demand for fresh agricultural products in Liaoning province.

| Agricultural products who need cold chain logistics | Fresh fruits and vegetables                                                                                                                                                                                                                                                                                    |                                                     |                                      |  | Aquatic product                                                                    | Eggs                                                                  | Milk                                                                                   | Meat                                                                           |                                                                                                                                                                           |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------|--|------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | Fruits and vegetables Suitable for low temperature storage and transportation                                                                                                                                                                                                                                  | Fruits and vegetables less sensitive to cold damage | Cold sensitive fruits and vegetables |  |                                                                                    |                                                                       |                                                                                        | Chilled fresh meat                                                             | Hot fresh meat                                                                                                                                                            |
| Temperature requirements (°C)                       | -1 - 5                                                                                                                                                                                                                                                                                                         | 5 - 8                                               | 8 - 12                               |  | 0                                                                                  | 0                                                                     | 2 - 6                                                                                  | 0 - 5                                                                          | Normal temperature                                                                                                                                                        |
| Cold chain logistics requirements                   | Post-harvest selection, pre-cooling, temperature-controlled transportation and warehousing                                                                                                                                                                                                                     |                                                     |                                      |  | Cooling or micro-freezing, temperature controlled transport and warehousing        | Pre-cooling, temperature-controlled transportation and warehousing    | Cooling, sterilization, filling, temperature controlled transportation and warehousing | Cooling after slaughter, temperature controlled transportation and warehousing | The whole circulation process does not carry on the temperature control, the microorganism cannot be restrained, easy to pollute, cannot guarantee the meat edible safety |
| Requirements for cold chain facilities              | Pre-cooling room, air-conditioned storage, temperature-controlled transport vehicles                                                                                                                                                                                                                           |                                                     |                                      |  | Cooling or micro-freezing equipment, cold storage, refrigerated transport vehicles | Pre-cooling room, temperature-controlled storage & transport vehicles | Pre-cooling room, temperature-controlled storage & transport vehicles                  | Pre-cooling room, cold storage, refrigerated transport vehicles                |                                                                                                                                                                           |
| Quality requirements for cold chain logistics       | Ensure temperature control throughout the flow process from harvest to delivery to consumers                                                                                                                                                                                                                   |                                                     |                                      |  |                                                                                    |                                                                       |                                                                                        |                                                                                |                                                                                                                                                                           |
| Capacity for cold chain logistics costs             | The affordability of cold chain logistics costs is generally low, except for a small number of high-value fruits and vegetables                                                                                                                                                                                |                                                     |                                      |  | Generally high                                                                     | Generally low                                                         | Generally low                                                                          | Medium                                                                         |                                                                                                                                                                           |
| Demand for cold chain logistics talent/skills       | Cold chain logistics grassroots operators and middle and senior management personnel who master logistics management, food quality management, refrigeration technology, refrigeration technology, information technology and other areas of knowledge and logistics talent/skills of the cold chain logistics |                                                     |                                      |  |                                                                                    |                                                                       |                                                                                        |                                                                                |                                                                                                                                                                           |

Figure 2 Cold chain logistics demand for fresh agricultural products in China

Analyzing Figure 2 **Error! Reference source not found.** presents the following characteristics:

- 1) Pre-cooling is the first step for fresh agricultural products to enter the cold chain system.
- 2) Fresh agricultural products circulation needs temperature control.
- 3) Fresh agricultural products have higher requirements for logistics equipment.
- 4) The circulation of fresh agricultural products needs the specialized personnel of cold chain logistics to operate and manage.

Having understood the overall cold-chain brief from above we consider a report from Shah Vatsall [6] from the International Research Journal of Engineering and Technology (IRJET), in which he unpacks the use of Raspberry Pi sensors.

The Raspberry Pi is a low cost, credit-card sized computer, it contains a sensor hat that can sense humidity and temperature [6]. It uses low energy consumption and is relatively small, which makes it suitable to fit in many containers that are designed for traveling [6]. Figure 3 below is a visual of Raspberry Pi.



Figure 3 Raspberry Pi

### 3. USER REQUIREMENTS

Below are some of the user requirements that outline the implementation of the Last-Mile Project.

**Product Details:** The user inserts parcel name, sender, and receiver details

**Temperature:** The user inserts desired temperature range of the product in the system.

**Pickup and Delivery Location Selection:** The user selects the pickup and delivery locations for the selected product.

**Collection:** The driver should be able to scan the product QR code of the product and upload its information to the system before departure.

**Real-time Tracking:** All stakeholders should be able to track the product in real-time, providing updates on the location, temperature, and estimated time of arrival.

**Alert Notifications:** The user and driver should receive notifications in case of any temperature fluctuations, breaches, or delays during transportation.

**On Delivery:** All stakeholders should have access to the recorded data of the product history.

#### 4. SYSTEM REQUIREMENTS

Below are some of the system requirements that outline the implementation of the Last-Mile Project.

**Temperature Control:** The system must ensure that the temperature of the products is within the required range during transportation. It should be able to adjust the temperature to the required range in case of fluctuations (future works).

**Product Tracking:** The system should allow all stakeholders to track their products in real-time, providing updates on the location, temperature, and estimated time of arrival.

**Packaging:** The system should ensure that the products are properly packaged and labelled to identify their temperature range and ensure proper handling during transportation.

**Dry Ice Management:** The system must be able to calculate the expected duration of the dry ice and provide alerts to the driver and customers when the ice needs to be refilled.

**Real-time Temperature Monitoring:** The system should monitor the temperature of the products in real-time during transportation and provide alerts to the driver and customers in case of temperature fluctuations or breaches.

**Driver Dispatch:** The system should be able to dispatch drivers depending on how far the driver is and how much ice they have available in the cooler box.

**Traceability:** The system allows drivers to upload product information to the system using QR codes. The uploaded data should be encrypted to avoid data modification and fraud.

**Dynamic Routing:** The system should use the shortest path algorithm explained in Section 8. It will help provide the shortest paths for drivers to reach their destinations in the minimal time possible.

**Device Compatibility:** The system should be designed to function in Android, Windows and IOS devices for different parts of the system.

**Data Analytics:** The system should be able to collect data on temperature fluctuations, transportation time, and other relevant metrics to analyse and optimize the cold chain logistics system for improved efficiency and cost-effectiveness.

#### 5. HARDWARE AND TOOLS

One of the important factors in the system is the cooler-box container used for the product storage. The SAIAMC will use a passive cooling container (cooler-box). The cooler-box will be kept at a certain temperature using a type of ice with a specific freezing point. In this project we have decided to choose the dry ice discussed in section 6.

**5.1. Raspberry Pi:** This microprocessor was considered as a possible solution due to its capabilities. These include light weight, low power consumption, easy sensor embedding and access to the internet. The disadvantage is the software is not simple for building basic input and output programs.



Figure 4 Raspberry Pi

**5.2. Arduino:** This is also an open-source product developed by an Italian company Arduino. It has similar capabilities as the raspberry pi, and it also provides a simpler software environment for simple input and output functions. The Arduino provides different models some from the original company and others are third parties.

For this project we have decided to use Arduino uno and Arduino (ESP8266MOD). The Arduino UNO is one of the original models built by Arduino. The most popular programming language used by Arduino is C++. This original model does not provide any additional features such as GSM or Wi-Fi these are embedded to it as shield. On the other hand, the esp8266 has a built in Wi-Fi drive that can be configured programmatically to connect to wireless networks [7].



Figure 5 Arduino ESP8266

Having underlined the similarities between the Arduino different microprocessors used for this project we now focus on the external sensors that read in the data from the external environment.

**GPS:** This module is connected to the Arduino board to receive GPS coordinates and send them to the microprocessor. -

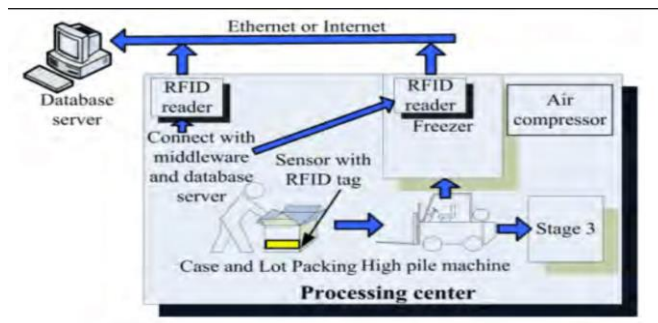


**Figure 6 NEO 6m GPS Module**

GSM: This module allows the Arduino microprocessor to use a sim card to connect to the internet via mobile data. This is the desired functionality because WIFI connections are not suitable for a constant moving device.

**Figure 7 GSM**

The RFID technology has been widely used for tracking products in different settings and businesses. Though it can be used for tracking cold chain products, we have adopted the QR Code technology because it seems more flexible and is becoming available and cheaper because of the availability on smart phones.

**Figure 8 RFID System Process**

## 6. SOFTWARE TOOLS

The administrator software where the organization will monitor the number of available drivers, number of orders and the data generated will be built using ReactJS frameworks [8]. According to the ReactJS framework, ReactJS is an open-source framework by Facebook for building web-based applications using JavaScript.

The customer and driver will be strictly mobile applications that will also be built using Flutter [9] this will assure that the application is compatible with both IOS and Android devices. The google map API will be integrated within the applications to provide live location updates. Figure 9 Shows the devices that are supported by Flutter.

**Figure 9 Flutter Devices**

## 7. SYSTEM DESIGN

In this Section we discuss the overall steps that are needed to successfully complete the Last-Mile delivery.

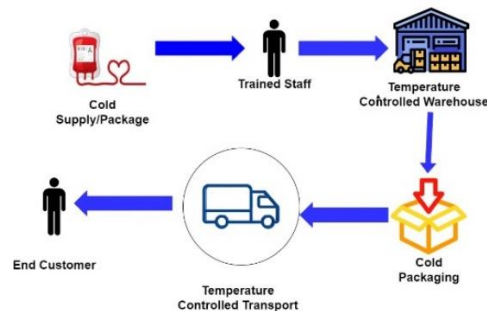
**Figure 10 Cold Chain Delivery Cycle**

Figure 10 illustrates the cold-chain delivery cycle where the following steps are used to deliver a cold Supply/package to the end customer.

### 1. Receive Order:

- a. When an order is created the admin will insert the order parameters, these include warehouse coordinates, destination coordinates, product temperature range.
- b. Finally, the order will be approved and sent to Mellow Cabs for driver dispatch. See **Error! Reference source not found.** illustrating a Mellow Cab



Figure 11 Mellow Cab vehicle

## 2. Dispatch:

- a. Mellow Cabs will dispatch a driver to collect the product. When the driver arrives at the warehouse, they must follow the collection procedure.

## 3. Collection:

- a. When the driver arrives at the warehouse, the initial step is to scan the product barcode using the last-mile mobile application discussed in section 6. The scanner will verify the product barcode and confirm the identity of the driver based on the application's details.
- b. When the driver collects the product, he/she will use a temperature sensor to sense and record the product temperature before inserting it inside the cooler-box. This data will be compared with the range that was inserted in step one. This will ensure that the product has not been compromised before the start of the trip.
- c. If the temperature of the product is not within the desired range of the temperature the order will be canceled otherwise it will proceed to the next stage (delivery).
- d. Once the driver has collected the product the system will generate the shortest path to reach the destination, this is discussed in Section 8.

## 4. On Route:

- a. When the driver departs the warehouse, the data coming from sensors embedded in the cooler box will send data to the local database and to the cloud. The driver will receive alerts and notifications on the App if there are any updates about the package.

## 5. Delivery:

- a. The data that has been stored throughout the delivery is made available to the client via the application. With this information, the client will be able to see the history of their

products' temperature to verify whether the quality of the product was compromised or not. This will be the final stage of the cold chain, as the product has reached its destination.

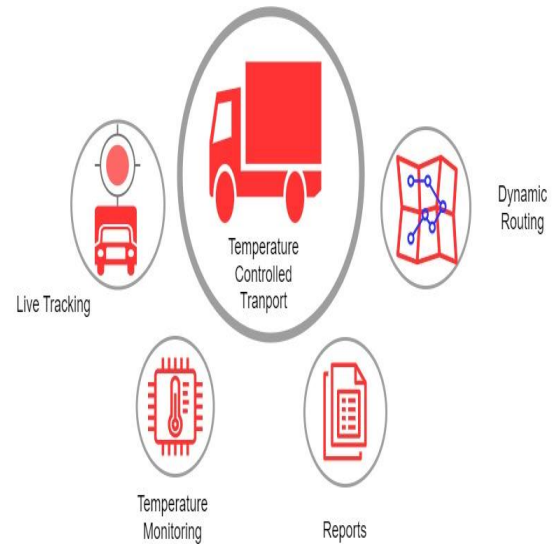


Figure 12 Cold Chain Delivery Processes

While Figure 10 illustrated the different steps of the cold chain delivery cycle, Figure 12 shows the key components involved in the cycle. After packet collection, they include.

- i) Dynamic routing of the cold chain vehicle
- ii) Live tracking of the package and vehicle
- iii) Cold package temperature monitoring and
- iv) reporting after delivery of the packet to the end user.

## 8. PICKUP AND DELIVERY OPTIMAZATION

The amount of dry ice left in the cooler box will hugely affect which path the driver must use, e.g., when there is more ice in the cooler box the driver can make many pick-ups and drop-offs without having to return to the warehouse for a refill.

As our transport vehicle has a constraint, which is the amount of dry ice, this calls for an optimal solution to dispatch drivers systematically to complete the orders. We consider a comparison between two systems TomTom API and Google OR Tools [10].

*Google OR Tools:* Google Optimization Tools (a.k.a., OR-Tools) is an open-source, fast and portable software suite for solving combinatorial optimization problems [11]. This software has the potential to generate optimized paths for different drivers to pick up and deliver orders by considering factors such as vehicle



constraints in our case amount of ice left. One of the challenges with the (GOT) is that it only works for problems who theoretically have solutions, the software needs major modification to incorporate scenarios where there is no solution. Another challenge is it depends on third party APIs to generate the distance between two coordinates on the map.

**TomTom API:** TomTom [12] is a Dutch company known for its navigation, mapping, and traffic services. They provide GPS devices, software, digital maps, and traffic information solutions widely used in automotive and mobile navigation systems. This API generates an optimized path based on live data for the driver to follow. Its challenge is that we do not have the ability to modify the algorithm used for path optimization as it is not an open source.

To conclude, based on the above router optimization techniques TomTom is considered as the choice of this project. This is because it uses live data, path optimization is consistent and does not rely on other third-party APIs.

## 9. DYNAMIC ROUTING

After the product is collected, it is important to consider the shortest path to the destination. Finding the shortest path is also viable to tell the driver which products can be delivered with the same trip. It is also important to consider factors such as traffic, road construction and accidents that might be present. The bellman ford and Dijkstra algorithm are considered as potential solutions to solve the problem [13].

Dijkstra's Algorithm works by maintaining a priority queue of vertices to visit, starting with the source vertex [13]. It visits each vertex adjacent to the current vertex, updating the distance to that vertex if it can be reached through a shorter path. The algorithm terminates when it reaches the destination vertex, or when all the reachable vertices have been visited. Dijkstra's Algorithm is commonly used in GPS navigation systems to find the shortest route between two points.

On the other hand, the Bellman-Ford Algorithm works by maintaining an array of distances from the source vertex to all other vertices in the graph [13]. It iteratively relaxes every edge in the graph, meaning that it checks if the distance to a vertex can be improved by taking a different path to that vertex. The algorithm repeats this process  $V - 1$  times (where  $V$  is the number of vertices in the graph) to ensure that all possible paths have been considered. The Bellman-Ford Algorithm is used when there are negative weights (i.e., costs or distances) in the graph, which Dijkstra's Algorithm does not handle well [13].

We also consider the TomTom API, which includes features such as shortest path finding and geo-locating coordinates. The TomTom [12] API is widely preferred by developers worldwide due to its dependable real-time traffic tracking and map

navigation capabilities. We have integrated it into this project for its multi-platform flexibility."

Based on the above comparisons, the TomTom API will be most suitable for the services of this application. To substantiate this, we understand that we don't only consider the distance between the locations but the time it will take to arrive there based on the factors (weight on vertices).

## 10. FLOW DIAGRAM

Figure 10 shows the flow diagram of the cold chain system.

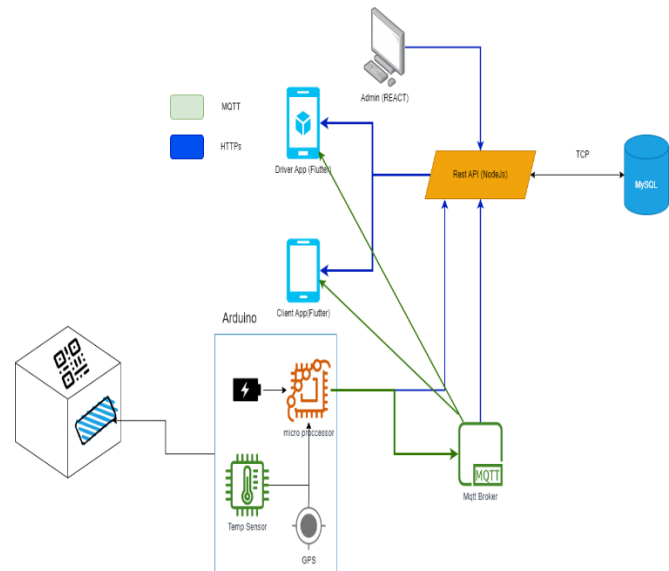


Figure 13 Flow Diagram

## 11. USE CASE DIAGRAM

Figure 14 is a visual representation of the system design discussed in Section 7.

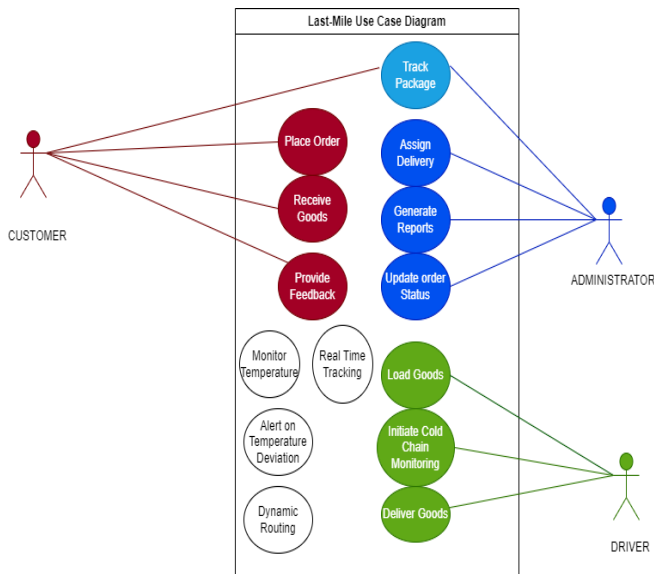


Figure 14 Use Case Diagram

## 12. PROTOTYPING

The prototype of this project is created using the following components.

### 1. HARDWARE

The hardware components are a physical mobile device or virtual device running IOS or android. A computer to host the server locally and an Arduino microprocessor to generate data.

### 2. SOFTWARE

The software components for the mobile applications included Figma which focused on mobile user interface design, draw.io to draw the UML, Functional diagrams and flow diagrams shown in Figure 10, Figure 11, and Figure 12.

## 13. IMPLEMENTAION

The implementation of the prototype uses flutter to shows the UI of both the client and driver applications.

## 14. HARDWARE

The requirements are physical or virtual emulator of an Android or IOS mobile device to run the application. A computer to act as the local server to host the database and API. The Arduino microprocessor, GPS module, and temperature sensor are attached to the container.

### 13.2 SOFTWARE

13.2.1 The front end is comprised of flutter (client and driver) and react (admin). Flutter is the development platform that brings everything together for the final mobile application look and feel. Flutter uses a

[TomTom](#) API to display the user location and order location on the map as markers. The mobile applications are connected to the server using the flutter [HTTP](#) package. This package is used to make http requests such as post, get update etc.

13.2.2 The backend is compromised of the server and the database. The server (API) connects the database to the frontend. The API is built with [NodeJS](#), within the NodeJS multiple libraries such as [express](#) is used for its simplicity to create endpoints. [Cors](#) is another package used to control permissions of the server e.g., to forbid certain IP addresses from accessing the server. The database is built using [MySQL](#) and connected to the sever using a NodeJS library called [Sequelize](#) which is an Object Relational Mapping (ORM).

13.2.3 Arduino program is written in C++ and uses different open source libraries for additional functionality, these being [TinyGPSPlus](#) for GPS, [WiFiManager](#) for Wi-Fi establishment. [PubSubClient](#) for MQTT connection to the broker. Below is a section of the Arduino code that imports the libraries.

```

#include <WiFiManager.h> //
#include <ArduinoMqttClient.h>
#include <TinyGPS++.h>
#include <ArduinoJson.h>
#include <string.h>
#include <SoftwareSerial.h>

TinyGPSPlus gps;
SoftwareSerial SerialGPS(4,5);
WiFiManager wifiManager;
WiFiClient wifiClient;
MqttClient mqttClient(wifiClient);

const char broker[] = "broker.hivemq.com";
int port = 1883;
String topic_str = "ima_test";

```

13.2.4 HiveMQTT is an online MQTT broker that Arduino will publish its live data to. The data being the live GPS, timestamp, and temperature. Below is a screenshot of the private cluster provided by HiveMQTT.

| Message                                                                                                  | Topic   |
|----------------------------------------------------------------------------------------------------------|---------|
| ["lat":33.9189,"lng":18.4232,"speed":10,"time":"2023-09-21T12:55:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.9675,"lng":18.4069,"speed":10,"time":"2023-09-21T12:50:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.967,"lng":18.4066,"speed":10,"time":"2023-09-21T12:45:00Z","temp":25,"mac_address":"1.1.1.1"]  | 1.1.1.1 |
| ["lat":33.9665,"lng":18.4063,"speed":10,"time":"2023-09-21T12:40:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.966,"lng":18.406,"speed":10,"time":"2023-09-21T12:35:00Z","temp":25,"mac_address":"1.1.1.1"]   | 1.1.1.1 |
| ["lat":33.9655,"lng":18.4057,"speed":10,"time":"2023-09-21T12:30:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.965,"lng":18.4054,"speed":10,"time":"2023-09-21T12:25:00Z","temp":25,"mac_address":"1.1.1.1"]  | 1.1.1.1 |
| ["lat":33.9645,"lng":18.4051,"speed":10,"time":"2023-09-21T12:20:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.964,"lng":18.4048,"speed":10,"time":"2023-09-21T12:15:00Z","temp":25,"mac_address":"1.1.1.1"]  | 1.1.1.1 |
| ["lat":33.9635,"lng":18.4045,"speed":10,"time":"2023-09-21T12:10:00Z","temp":25,"mac_address":"1.1.1.1"] | 1.1.1.1 |
| ["lat":33.963,"lng":18.4042,"speed":10,"time":"2023-09-21T12:05:00Z","temp":25,"mac_address":"1.1.1.1"]  | 1.1.1.1 |

## 15. FUNCTIONS AND CLASSES

The last mile cold chain uses a variety of functions to achieve its goal of real time GPS and temperature monitoring.

The system is designed to accommodate three distinct user roles: clients, drivers, and administrators. Each of these user types shares common functionalities such as registration, login, and logout.

For clients, specialized functions have been implemented to enhance their experience. These include the ability to *create orders*, *cancel existing orders*, *track the progress of their orders*, and *update order details* as needed.

Drivers, on the other hand, have their own set of unique functions tailored to their role in the system. These functionalities encompass picking up assigned orders, delivering orders to their intended destinations, and promptly updating the status of orders to reflect the current action being taken.

Administrators, as overseers of the system, possess a distinct set of functions focused on managing the overall operations. This involves the approval of orders, allocation of drivers to pending orders, and the ability to track the real-time location of drivers for efficient coordination.

In summary, the system caters to the needs of clients, drivers, and administrators, ensuring a seamless and efficient experience for all parties involved. Whether it's creating, tracking, or updating orders, the system is equipped to handle a diverse range of tasks, tailored to the specific roles and responsibilities of each user type.

## 16. DATABASE CLASSES (UML)

Below is the UML diagram representing the structure of the database and the connection between the tables.

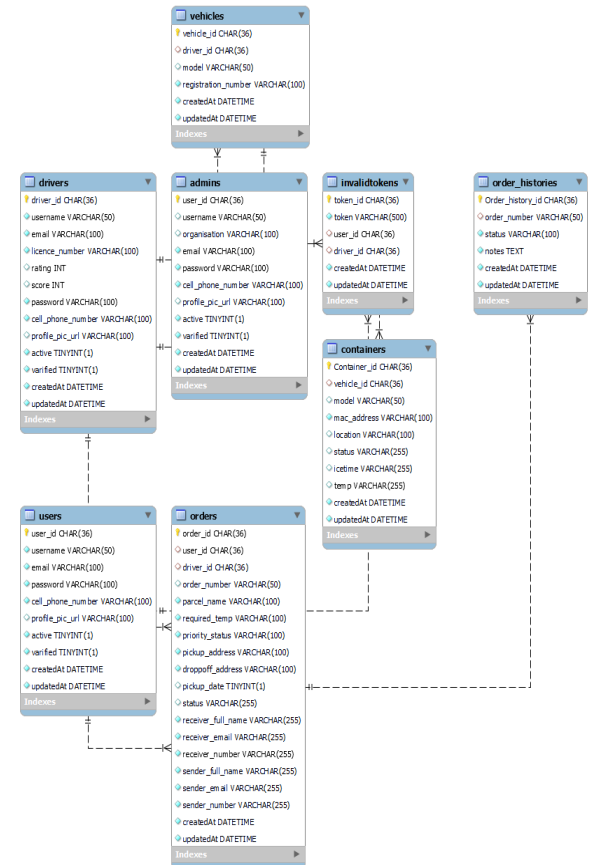


Figure 15 UML Diagram

## 17. USER INTERFACES FOR DIFFERENT SYSTEMS

### Orders Table

| # | Order Name        | Order Number | Origin                        | Destination                     | km/hr      | Time/minutes | Driver | Status | Action |
|---|-------------------|--------------|-------------------------------|---------------------------------|------------|--------------|--------|--------|--------|
| 0 | Vidae Caffe 2     | 76752027     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |
| 1 | genesis           | e893a134     | -33.9339516, 18.6279415       | -25.7825004, 27.865918599999999 | 1436.77 km | 1084.62 m    | Driver | BOOKED | Accept |
| 2 | Uwc Building      | 0f4c2574     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |
| 3 | Meal              | d7f37322     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |
| 4 | Vidae Caffe 2     | e02529f5     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |
| 5 | Vidae Caffe 2     | 2e1ab885     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |
| 6 | Uwc Swimming Pool | 23332c84     | -33.9319657686, 18.6264732559 | -33.9306532293, 18.6298257331   | 0.69 km    | 2.65 m       | Driver | BOOKED | Accept |

Figure 16 Admin Application



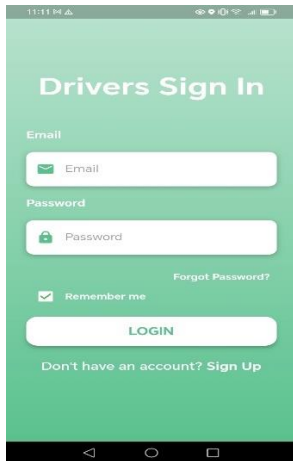


Figure 17 Driver

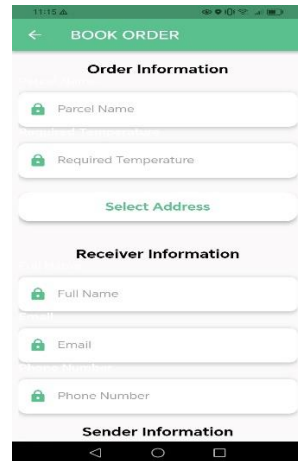


Figure 18 Client

## 18. TESTING: ARDUINO GPS AND TEMPERATURE DATA TRANSMISSION

The objective of this testing phase was to validate the functionality of the system in sending live GPS data and temperature readings from an Arduino device to an MQTT broker (Hivemq) hosted on a private cluster. Additionally, the Arduino was tasked with sending this data to a REST API at 5-minute intervals.

## 19. HARDWARE COMPONENTS

- Arduino device with GPS and temperature sensor modules
- Stable power source
- Computer for monitoring and debugging.

## 20. SOFTWARE COMPONENTS

- Arduino IDE for programming the device.
- MQTT client library for Arduino
- REST API endpoint for data reception
- Hivemq MQTT broker on the private cluster

## 21. TEST STEPS

### 22. Arduino Code Configuration:

- Developed and uploaded Arduino code for data acquisition from GPS and temperature sensors.
- Integrated MQTT client libraries to establish a connection with the Hivemq broker.
- Implemented logic for data transmission at 5-minute intervals to the REST API.

### 23. Hivemq Broker Configuration:

- Configured a private cluster to host the Hivemq MQTT broker.
- Generated access credentials and ensured secure communication channels.

### 24. Testing Live GPS Data Transmission:

- Activated the Arduino device and monitored the serial output for GPS data.
- Verified that live GPS coordinates were being acquired accurately.

### 25. Testing Live Temperature Data Transmission:

- Checked the Arduino serial output for temperature readings.
- Ensured that temperature data was captured correctly from the sensor.

### 26. Testing MQTT Data Transmission:

- Established a connection between the Arduino and the Hivemq broker.
- Verified that live GPS and temperature data were successfully published to respective MQTT topics.

### 27. Testing REST API Integration:

- Configured the REST API endpoint for receiving data from the Arduino.
- Verified that data was being received at 5-minute intervals as expected.

## 28. TEST RESULTS

**GPS Data Transmission:** GPS coordinates were consistently transmitted and received accurately.

**Temperature Data Transmission:** Live temperature readings were successfully transmitted and received without anomalies.

**MQTT Data Transmission:** Data was reliably published to the specified MQTT topics on the Hivemq broker.

**REST API Integration:** Data was received at 5-minute intervals, conforming to the specified requirement.

The testing phase was successful in validating the functionality of the system. The Arduino device effectively transmitted live GPS and temperature data to the Hivemq MQTT broker hosted on the private cluster. Additionally, the device adhered to the 5-minute data transmission intervals to the REST API. This confirms the system's readiness for deployment in a real-world environment.

## 29. CONCLUSIONS

This project addresses critical challenges within the last mile delivery process, emphasizing sustainability, process complexity, and evolving customer demands. Leveraging Fourth Industrial Revolution (4IR) technologies such as the Internet of Things, software engineering, and machine learning, the project focuses on developing and monitoring a hardware solution—a cooler box—to enable E-delivery vehicle manufacturers to seamlessly integrate into the cold chain delivery business.

The study delves into the realm of cold chain logistics, a specialized branch of logistics centered around refrigeration technology. It caters to products requiring low-temperature storage and transportation throughout their lifecycle, from production to sales. This network ensures product quality is preserved and minimizes product loss. To achieve this, a range of hardware tools like refrigerators, passive cooling containers, temperature sensors, humidity sensors, and light sensors are employed. Complementing these, software tools play a crucial role in integrating hardware components into a cohesive cold chain system.

The heart of this project, termed the "last-mile," focuses on designing and overseeing the hardware of a cooler box, enabling E-delivery vehicle manufacturers to partake in the cold chain delivery business. The last-mile system encompasses automatic driver dispatching, package collection, dynamic routing, and flexible product delivery facilitated by battery-powered or passive cooler boxes.

Future Recommendations:

1. **Extended Monitoring:** Observe the system's performance over an extended period to ensure sustained effectiveness under varying conditions.
2. **Load and Stress Testing:** Conduct comprehensive load and stress testing to evaluate system scalability and robustness.
3. **Error Handling Mechanisms:** Implement robust error-handling mechanisms to address potential network disruptions or data transmission failures.

This project represents a significant step forward in revolutionizing last-mile delivery, offering a comprehensive solution to meet the evolving demands of modern logistics. The integration of cutting-edge technology and thoughtful system design promises to enhance efficiency and reliability in cold chain logistics, contributing to the growth of the global logistics industry.

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